Отчет по практической работе №2

Студент: KAI

Группа: 16

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1. Подготовка к Лабораторной

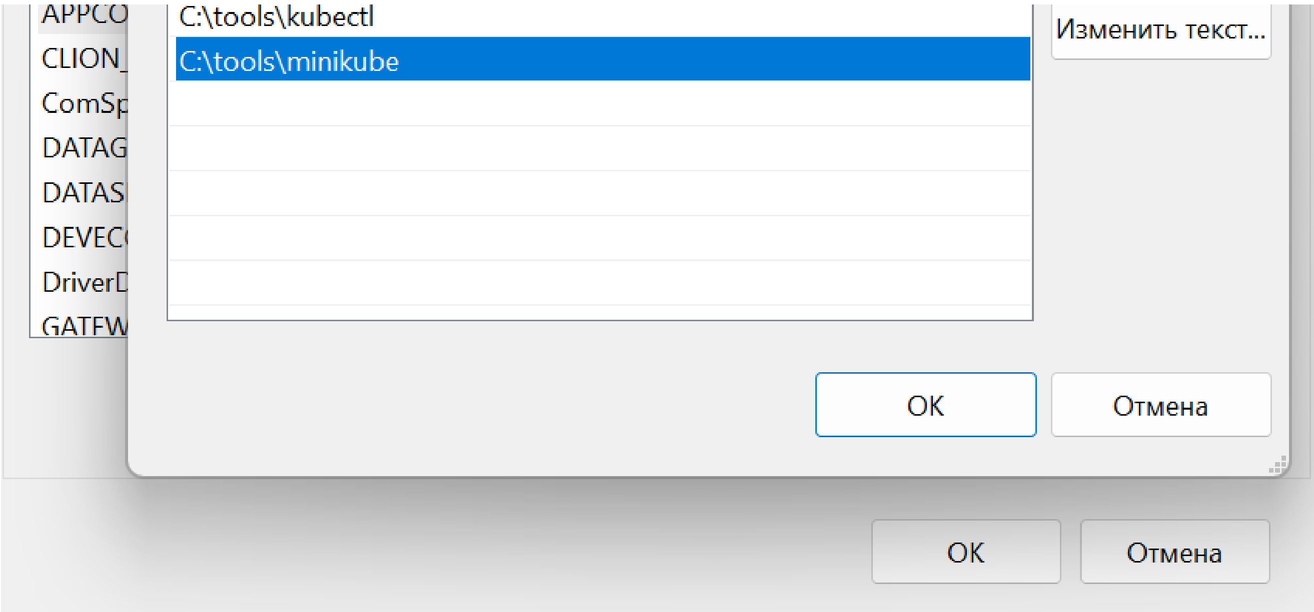
1.1 Установка kubectl

```
C:\Windows\System32>curl.exe -LO"https://dl.k8s.io/release/v1.28.0/bin/windows/amd64/kubectl.exe"
curl: (2) no URL specified
curl: try 'curl --help' for more information

C:\Windows\System32>curl.exe -LO "https://dl.k8s.io/release/v1.28.0/bin/windows/amd64/kubectl.exe"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 48.23M 100 48.23M   0      0 847.1k      0  00:58   00:58          1.02M

C:\Windows\System32>mkdir C:\tools\kubectlmove kubectl.exe C:\tools\kubectl\
Подпапка или файл kubectl.exe уже существует.
Ошибка во время обработки: kubectl.exe.

C:\Windows\System32>mkdir C:\tools\kubectl
Подпапка или файл C:\tools\kubectl уже существует.

C:\Windows\System32>move kubectl.exe C:\tools\kubectl\
Перемещено файлов:         1.

C:\Windows\System32>kubectl version --client
Client Version: v1.31.4
Kustomize Version: v5.4.2
```

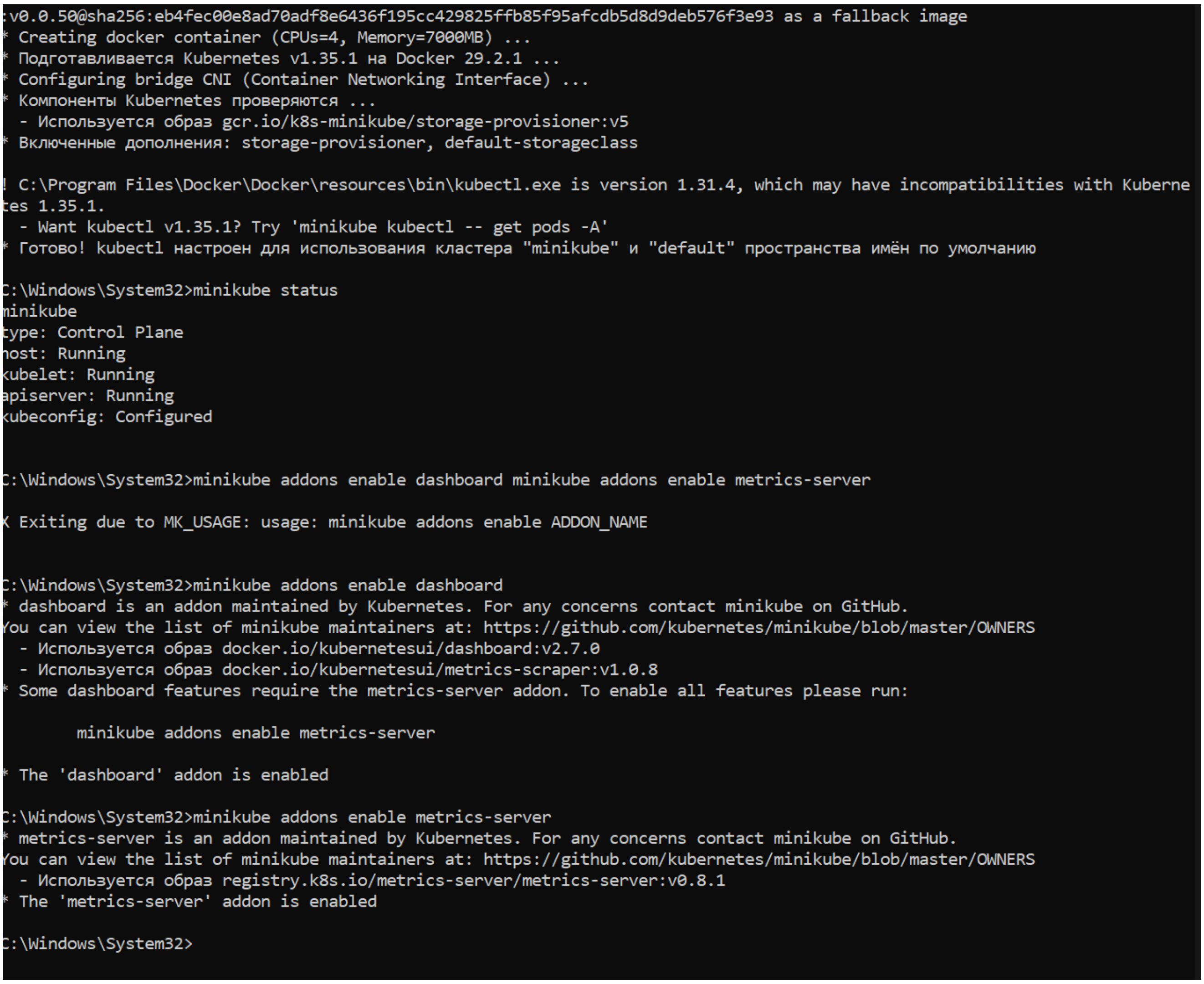
1.2 Установка Minikube

```
C:\Windows\System32>curl.exe -LO https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.
exe
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0      0   0      0   0      0      0      0                              0
  0      0   0      0   0      0      0      0           00:01              0
100 130.0M 100 130.0M   0      0 665.9k      0  03:19   03:19          2.20M

C:\Windows\System32>rename minikube-windows-amd64.exe minikube.exe
Файл с таким именем уже существует
или не найден.

C:\Windows\System32>mkdir C:\tools\minikube

C:\Windows\System32>move minikube.exe C:\tools\minikube\
Перемещено файлов:         1.

C:\Windows\System32>
```

1.3 Добавление в PATH



1.4 Запуск Minikube с драйвером Docker



```
:v0.0.50@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93 as a fallback image
* Creating docker container (CPUs=4, Memory=7000MB) ...
* Подготавливается Kubernetes v1.35.1 на Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Компоненты Kubernetes проверяются ...
  - Используется образ gcr.io/k8s-minikube/storage-provisioner:v5
* Включенные дополнения: storage-provisioner, default-storageclass

! C:\Program Files\Docker\Docker\resources\bin\kubectl.exe is version 1.31.4, which may have incompatibilities with Kuberne
tes 1.35.1.
  - Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
* Готово! kubectl настроен для использования кластера "minikube" и "default" пространства имён по умолчанию

C:\Windows\System32>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Windows\System32>minikube addons enable dashboard minikube addons enable metrics-server

X Exiting due to MK_USAGE: usage: minikube addons enable ADDON_NAME


C:\Windows\System32>minikube addons enable dashboard
* dashboard is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  - Используется образ docker.io/kubernetesui/dashboard:v2.7.0
  - Используется образ docker.io/kubernetesui/metrics-scraper:v1.0.8
* Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

* The 'dashboard' addon is enabled

C:\Windows\System32>minikube addons enable metrics-server
* metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  - Используется образ registry.k8s.io/metrics-server/metrics-server:v0.8.1
* The 'metrics-server' addon is enabled

C:\Windows\System32>
```

```
minikube status
type: Control Plane
host: Running
```

```
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

1.5 Настройка доступа к GHCR

```
secret/ghcr-secret created

C:\Windows\System32>kubectl get secrets
NAME          TYPE                             DATA   AGE
ghcr-secret   kubernetes.io/dockerconfigjson   1      33s

C:\Windows\System32>_
```

1.6 Подготовка репозитория

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (main)
$ git checkout -b lab2/kubernetes-basics
Switched to a new branch 'lab2/kubernetes-basics'

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ mkdir k8s

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ cd k8s/

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

Часть 2. Знакомство с kubectl и базовыми концепциями

2.1 Первые команды kubectl

Задание 2.1.1: Исследуйте кластер

```
C:\Windows\System32>kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:52074
CoreDNS is running at https://127.0.0.1:52074/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

C:\Windows\System32>kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   83m   v1.35.1

C:\Windows\System32>kubectl get nodes -o wide
NAME       STATUS   ROLES           AGE   VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE                         KERNEL-VE
RSION                       CONTAINER-RUNTIME
minikube   Ready    control-plane   83m   v1.35.1   192.168.49.2   <none>        Debian GNU/Linux 12 (bookworm)   5.15.167.
4-microsoft-standard-WSL2   docker://29.2.1

C:\Windows\System32>kubectl describe nodes
Name:               minikube
Roles:              control-plane
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    kubernetes.io/arch=amd64
                    kubernetes.io/hostname=minikube
                    kubernetes.io/os=linux
                    minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
                    minikube.k8s.io/name=minikube
                    minikube.k8s.io/primary=true
                    minikube.k8s.io/updated_at=2026_03_31T21_35_16_0700
                    minikube.k8s.io/version=v1.38.1
                    node-role.kubernetes.io/control-plane=
                    node.kubernetes.io/exclude-from-external-load-balancers=
Annotations:        node.alpha.kubernetes.io/ttl: 0
                    volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp:  Tue, 31 Mar 2026 21:35:11 +0300
Taints:             <none>
Unschedulable:      false
Lease:
  HolderIdentity:  minikube
  AcquireTime:     <unset>
  RenewTime:       Tue, 31 Mar 2026 22:58:53 +0300
Conditions:
  Type             Status  LastHeartbeatTime                 LastTransitionTime                Reason
 Message
  ----             ------  -----------------                 ------------------                ------
 -------
  MemoryPressure   False   Tue, 31 Mar 2026 22:54:23 +0300   Tue, 31 Mar 2026 21:35:09 +0300   KubeletHasSufficientMemory
```

```
C:\Windows\System32>kubectl api-resources | findstr pod
pods                              po           v1                         true         Pod
podtemplates                                   v1                         true         PodTemplate
horizontalpodautoscalers          hpa          autoscaling/v2             true         HorizontalPodAutoscaler
pods                                           metrics.k8s.io/v1beta1     true         PodMetrics
poddisruptionbudgets              pdb          policy/v1                  true         PodDisruptionBudget

C:\Windows\System32>kubectl api-resources | findstr deployment
deployments                       deploy       apps/v1                    true         Deployment

C:\Windows\System32>
```

```
- cluster:
    certificate-authority-data: DATA+OMITTED
    server: https://127.0.0.1:6443
  name: docker-desktop
- cluster:
    certificate-authority: C:\Users\Aleksandr\.minikube\ca.crt
    extensions:
    - extension:
        last-update: Tue, 31 Mar 2026 21:35:17 MSK
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: cluster_info
    server: https://127.0.0.1:52074
  name: minikube
contexts:
- context:
    cluster: docker-desktop
    user: docker-desktop
  name: docker-desktop
- context:
    cluster: minikube
    extensions:
    - extension:
        last-update: Tue, 31 Mar 2026 21:35:17 MSK
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: context_info
    namespace: default
    user: minikube
  name: minikube
current-context: minikube
kind: Config
preferences: {}
users:
- name: docker-desktop
  user:
    client-certificate-data: DATA+OMITTED
    client-key-data: DATA+OMITTED
- name: minikube
  user:
    client-certificate: C:\Users\Aleksandr\.minikube\profiles\minikube\client.crt
    client-key: C:\Users\Aleksandr\.minikube\profiles\minikube\client.key

C:\Windows\System32>kubectl config current-context
minikube

C:\Windows\System32>
```

Самостоятельное задание (nodes.txt):

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes -o wide > nodes.txt
```

```
NAME       STATUS   ROLES           AGE   VERSION   INTERNAL-IP    EXTERNAL-IP
OS-IMAGE                         KERNEL-VERSION
CONTAINER-RUNTIME
minikube   Ready    control-plane   22h   v1.35.1   192.168.49.2   <none>
Debian GNU/Linux 12 (bookworm)   5.15.167.4-microsoft-standard-WSL2
docker://29.2.1
```

2.2 Работа с подами (Pods)

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl apply -f k8s/01-pod-nginx.yaml

pod/nginx-pod created

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods
NAME        READY   STATUS              RESTARTS   AGE
nginx-pod   0/1     ContainerCreating   0          8s

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED NODE   READINESS GATES
nginx-pod   1/1     Running   0          24s   10.244.0.6   minikube   <none>           <none>

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods --show-labels
NAME        READY   STATUS    RESTARTS   AGE   LABELS
nginx-pod   1/1     Running   0          29s   app=nginx,lab=lab2

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl describe pod nginx-pod
Name:             nginx-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 31 Mar 2026 23:05:26 +0300
Labels:           app=nginx
                  lab=lab2
Annotations:      <none>
Status:           Running
IP:               10.244.0.6
IPs:
  IP:  10.244.0.6
Containers:
  nginx:
    Container ID:   docker://94ab0e8f6f7dec84aa4f04502a0897fc6744c24199ccd0b790ca46de50191c92
    Image:          nginx:alpine
    Image ID:       docker-pullable://nginx@sha256:e7257f1ef28ba17cf7c248cb8ccf6f0c6e0228ab9c315c152f9c203cd34cf6d1
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Tue, 31 Mar 2026 23:05:41 +0300
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:        250m
```

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl logs nginx-pod
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/31 20:05:41 [notice] 1#1: using the "epoll" event method
2026/03/31 20:05:41 [notice] 1#1: nginx/1.29.7
2026/03/31 20:05:41 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/31 20:05:41 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2026/03/31 20:05:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/31 20:05:41 [notice] 1#1: start worker processes
2026/03/31 20:05:41 [notice] 1#1: start worker process 30
2026/03/31 20:05:41 [notice] 1#1: start worker process 31
2026/03/31 20:05:41 [notice] 1#1: start worker process 32
2026/03/31 20:05:41 [notice] 1#1: start worker process 33
2026/03/31 20:05:41 [notice] 1#1: start worker process 34
2026/03/31 20:05:41 [notice] 1#1: start worker process 35
2026/03/31 20:05:41 [notice] 1#1: start worker process 36
2026/03/31 20:05:41 [notice] 1#1: start worker process 37
2026/03/31 20:05:41 [notice] 1#1: start worker process 38
2026/03/31 20:05:41 [notice] 1#1: start worker process 39
2026/03/31 20:05:41 [notice] 1#1: start worker process 40
2026/03/31 20:05:41 [notice] 1#1: start worker process 41

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl logs nginx-pod -f
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/31 20:05:41 [notice] 1#1: using the "epoll" event method
```

```
2026/03/31 20:05:41 [notice] 1#1: nginx/1.29.7
2026/03/31 20:05:41 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/31 20:05:41 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2026/03/31 20:05:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/31 20:05:41 [notice] 1#1: start worker processes
2026/03/31 20:05:41 [notice] 1#1: start worker process 30
```

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl exec -it nginx-pod -- sh
/ # curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>
```

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl top pod nginx-pod
NAME        CPU(cores)   MEMORY(bytes)
nginx-pod   0m           10Mi

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl delete pod nginx-pod
pod "nginx-pod" deleted

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>
```

Самостоятельное задание (Go-приложение из ЛР1):

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl apply -f k8s/pod_test_practic.
yaml
pod/go-lr1-pod created

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods
NAME         READY   STATUS              RESTARTS   AGE
go-lr1-pod   0/1     ContainerCreating   0          6s

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl logs go-lr1-pod > crash-logs.t
xt

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>
```

```
2026/03/31 20:14:24 Error creating table:dial tcp [::1]:5432: connect:
connection refused
```

2.3 Работа с ReplicaSet

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl apply -f k8s/02-replicaset-ngi
nx.yaml
replicaset.apps/nginx-replicaset created

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get replicasets
NAME               DESIRED   CURRENT   READY   AGE
nginx-replicaset   3         3         3       7s

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get rs
NAME               DESIRED   CURRENT   READY   AGE
nginx-replicaset   3         3         3       12s

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl describe rs nginx-replicaset
Name:         nginx-replicaset
Namespace:    default
Selector:     app=nginx
Labels:       app=nginx
              lab=lab2
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:         nginx:alpine
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Events:
  Type    Reason            Age   From                   Message
  ----    ------            ----  ----                   -------
  Normal  SuccessfulCreate  19s   replicaset-controller  Created pod: nginx-replicaset-9j57c
  Normal  SuccessfulCreate  19s   replicaset-controller  Created pod: nginx-replicaset-rb22q
  Normal  SuccessfulCreate  19s   replicaset-controller  Created pod: nginx-replicaset-prc6q

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>
```

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods -l app=nginx
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-9j57c   1/1     Running   0          40s
nginx-replicaset-prc6q   1/1     Running   0          40s
nginx-replicaset-rb22q   1/1     Running   0          40s

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl scale rs nginx-replicaset --re
plicas=5
replicaset.apps/nginx-replicaset scaled

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods -l app=nginx -w
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-762tp   1/1     Running   0          7s
nginx-replicaset-9j57c   1/1     Running   0          55s
nginx-replicaset-prc6q   1/1     Running   0          55s
nginx-replicaset-rb22q   1/1     Running   0          55s
nginx-replicaset-rlfch   1/1     Running   0          7s
```

```
C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl delete pod nginx-replicaset-76
2tp
pod "nginx-replicaset-762tp" deleted

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>kubectl get pods -l app=nginx
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-2nmdf   1/1     Running   0          11s
nginx-replicaset-9j57c   1/1     Running   0          108s
nginx-replicaset-prc6q   1/1     Running   0          108s
nginx-replicaset-rb22q   1/1     Running   0          108s
nginx-replicaset-rlfch   1/1     Running   0          60s

C:\Users\Aleksandr\Desktop\subjects\Предметы\2026 Docker Докер\lab1\containers-lab-1>
```

2.4 Работа с Deployment

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-app.yaml
deployment.apps/go-app-deployment created

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
go-app-deployment   0/2     2            0           24s

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe deployment go-app-deployment
Name:                   go-app-deployment
Namespace:              default
CreationTimestamp:      Wed, 01 Apr 2026 13:33:16 +0300
Labels:                 app=go-app
                        lab=lab2
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=go-app
Replicas:               2 desired | 2 updated | 2 total | 0 available | 2 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  0 max unavailable, 1 max surge
Pod Template:
  Labels:  app=go-app
  Containers:
   go-app:
    Image:      ghcr.io/nightfury2283/containers-lab-1/app:sha-c94b5e6
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get rs -l app=go-app
NAME                          DESIRED   CURRENT   READY   AGE
go-app-deployment-85fbb9ff74   2         2         0       2m59s

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=go-app
NAME                                 READY   STATUS             RESTARTS      AGE
go-app-deployment-85fbb9ff74-b8896   0/1     CrashLoopBackOff   4 (42s ago)   3m7s
go-app-deployment-85fbb9ff74-d58kd   0/1     CrashLoopBackOff   4 (42s ago)   3m7s

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl rollout history deployment go-app-deployment
deployment.apps/go-app-deployment
REVISION  CHANGE-CAUSE
1         <none>


Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale deployment go-app-deployment --replicas=3
deployment.apps/go-app-deployment scaled
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl set image deployment/go-app-deployment go-app=ghcr.io/nightfury2283/containers-lab-1/app:new-tag
deployment.apps/go-app-deployment image updated
```

Самостоятельное задание (PostgreSQL Deployment):

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl apply -f test-deployment-postgres.yaml
deployment.apps/postgres-deployment created

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get pods
NAME                                  READY   STATUS              RESTARTS        AGE
go-app-deployment-56f576955c-cz6ns    0/1     InvalidImageName    0               6m31s
go-app-deployment-75bbc799c6-655x6    0/1     ImagePullBackOff    0               4m38s
go-app-deployment-85fbb9ff74-d58kd    0/1     CrashLoopBackOff    6 (3m43s ago)   10m
go-app-deployment-85fbb9ff74-rvghb    0/1     CrashLoopBackOff    6 (28s ago)     7m20s
postgres-deployment-5bcd58fdd7-kx5wv  0/1     ContainerCreating   0               2s
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl logs -l app=go-app
2026/04/01 10:49:10 Error creating table:dial tcp: lookup postgres-service on 10.96.0.10:53: server misbehaving
```

2.5 Работа с Service

```
04-service-postgres.yaml    U
05-service-app.yaml         U
06-service-nginx.yaml       U
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/04-service-postgres.yaml
service/postgres-service created

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/05-service-app.yaml
service/go-app-service created

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/06-service-nginx.yaml
service/nginx-service created

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get services
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
go-app-service     ClusterIP   10.110.49.120    <none>        8080/TCP       20s
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        16h
nginx-service      NodePort    10.111.125.143   <none>        80:30080/TCP   10s
postgres-service   ClusterIP   10.108.246.120   <none>        5432/TCP       38s

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe svc postgres-service
Name:                     postgres-service
Namespace:                default
Labels:                   app=postgres
                          lab=lab2
Annotations:              <none>
Selector:                 app=postgres
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.108.246.120
IPs:                      10.108.246.120
Port:                     <unset>  5432/TCP
TargetPort:               5432/TCP
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl run test-pod --image=alpine:latest --rm -it --restart=Never -- sh
If you don't see a command prompt, try pressing enter.
/ # apk add curl postgresql-client
( 1/17) Installing brotli-libs (1.2.0-r0)
( 2/17) Installing c-ares (1.34.6-r0)
( 3/17) Installing libunistring (1.4.1-r0)
( 4/17) Installing libidn2 (2.3.8-r0)
( 5/17) Installing nghttp2-libs (1.68.0-r0)
( 6/17) Installing nghttp3 (1.13.1-r0)
( 7/17) Installing libpsl (0.21.5-r3)
( 8/17) Installing zstd-libs (1.5.7-r2)
( 9/17) Installing libcurl (8.17.0-r1)
(10/17) Installing curl (8.17.0-r1)
(11/17) Installing postgresql-common (1.2-r2)
  Executing postgresql-common-1.2-r2.pre-install
(12/17) Installing lz4-libs (1.10.0-r0)
(13/17) Installing libpq (18.3-r0)
(14/17) Installing ncurses-terminfo-base (6.5_p20251123-r0)
(15/17) Installing libncursesw (6.5_p20251123-r0)
(16/17) Installing readline (8.3.1-r0)
(17/17) Installing postgresql18-client (18.3-r0)
Executing busybox-1.37.0-r30.trigger
Executing postgresql-common-1.2-r2.trigger
* Setting postgresql18 as the default version
OK: 17.6 MiB in 33 packages
/ # curl go-app-service:8080/health
curl: (7) Failed to connect to go-app-service port 8080 after 4 ms: Could not connect to server
/ # pg_isready -h postgres-service
postgres-service:5432 - accepting connections
/ # exit
pod "test-pod" deleted
```

2.6 Полный стек приложения

Часть 3. Валидация и отладка манифестов

3.1 Валидация YAML

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-app.yaml --dry-run=client
deployment.apps/go-app-deployment configured (dry run)

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain pod
KIND:       Pod
VERSION:    v1

DESCRIPTION:
    Pod is a collection of containers that can run on a host. This resource is
    created by clients and scheduled onto hosts.

FIELDS:
  apiVersion    <string>
    APIVersion defines the versioned schema of this representation of an object.
    Servers should convert recognized schemas to the latest internal value, and
    may reject unrecognized values. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

  kind  <string>
    Kind is a string value representing the REST resource this object
    represents. Servers may infer this from the endpoint the client submits
    requests to. Cannot be updated. In CamelCase. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

  metadata      <ObjectMeta>
    Standard object's metadata. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata

  spec  <PodSpec>
    Specification of the desired behavior of the pod. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain deployment.spec
GROUP:      apps
KIND:       Deployment
VERSION:    v1

FIELD: spec <DeploymentSpec>


DESCRIPTION:
    Specification of the desired behavior of the Deployment.
    DeploymentSpec is the specification of the desired behavior of the
    Deployment.

FIELDS:
  minReadySeconds       <integer>
    Minimum number of seconds for which a newly created pod should be ready
    without any of its container crashing, for it to be considered available.
    Defaults to 0 (pod will be considered available as soon as it is ready)

  paused        <boolean>
    Indicates that the deployment is paused.

  progressDeadlineSeconds       <integer>
    The maximum time in seconds for a deployment to make progress before it is
    considered to be failed. The deployment controller will continue to process
    failed deployments and a condition with a ProgressDeadlineExceeded reason
    will be surfaced in the deployment status. Note that progress will not be
    estimated during the time a deployment is paused. Defaults to 600s.

  replicas      <integer>
    Number of desired pods. This is a pointer to distinguish between explicit
    zero and not specified. Defaults to 1.
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain deployment.spec.template.spec.containers
GROUP:      apps
KIND:       Deployment
VERSION:    v1

FIELD: containers <[]Container>


DESCRIPTION:
    List of containers belonging to the pod. Containers cannot currently be
    added or removed. There must be at least one container in a Pod. Cannot be
    updated.
    A single application container that you want to run within a pod.

FIELDS:
  args  <[]string>
    Arguments to the entrypoint. The container image's CMD is used if this is
    not provided. Variable references $(VAR_NAME) are expanded using the
    container's environment. If a variable cannot be resolved, the reference in
    the input string will be unchanged. Double $$ are reduced to a single $,
    which allows for escaping the $(VAR_NAME) syntax: i.e. "$$(VAR_NAME)" will
```

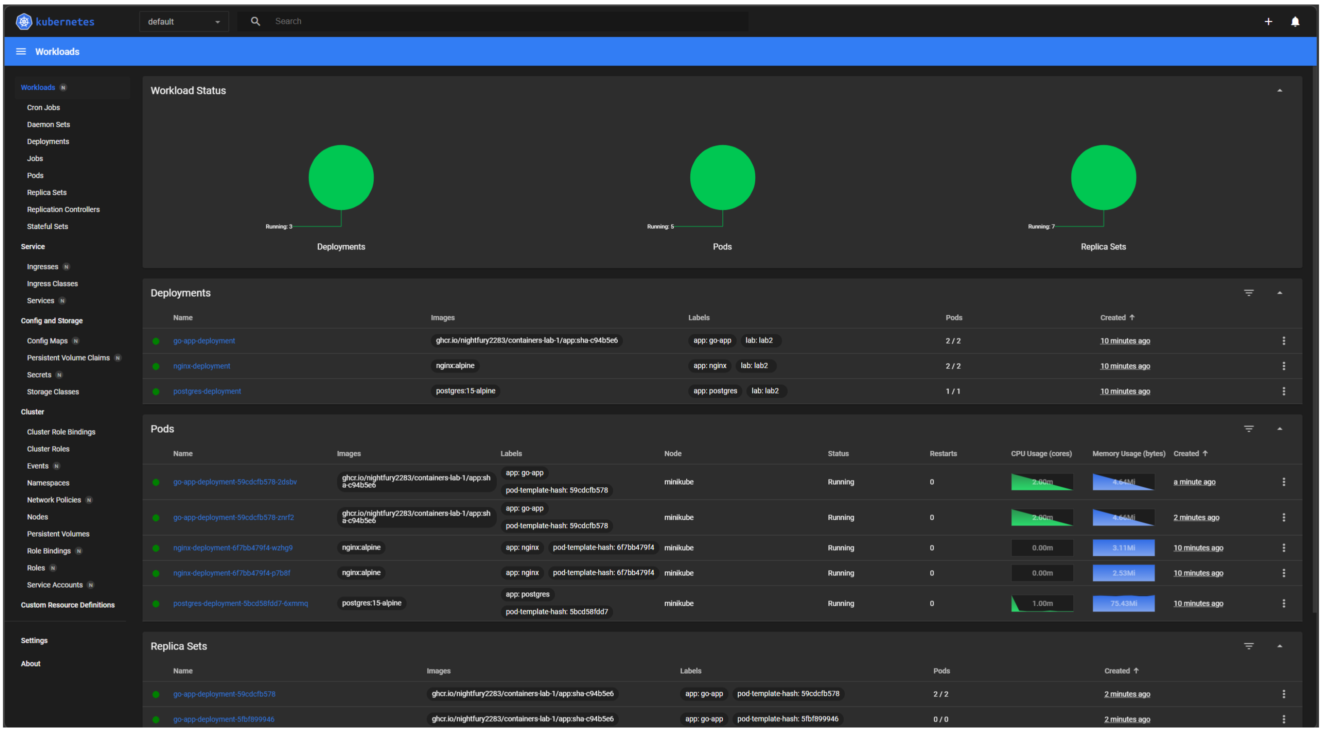
3.2 Отладка приложений

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get events --sort-by='.lastTimestamp'
LAST SEEN   TYPE      REASON              OBJECT                                    MESSAGE
59m         Normal    Killing             pod/go-app-deployment-85fbb9ff74-dxc7b    Container go-app failed liveness probe, will be restarted
49m         Warning   BackOff             pod/go-app-deployment-85fbb9ff74-xtz94    Back-off restarting failed container go-app in pod go-app-deployment-85fbb9ff74-xtz94_default(1dad4b
b-c6a0-4a6e-844e-bf915df2723a)
48m         Warning   BackOff             pod/go-app-deployment-85fbb9ff74-hr2c7    Back-off restarting failed container go-app in pod go-app-deployment-85fbb9ff74-hr2c7_default(c4856b
d-7b40-476a-9711-80af947f16a8)
46m         Normal    Pulled              pod/go-app-deployment-85fbb9ff74-hr2c7    Container image "ghcr.io/nightfury2283/containers-lab-1/app:sha-c94b5e6" already present on machine
nd can be accessed by the pod
44m         Warning   Unhealthy           pod/go-app-deployment-85fbb9ff74-xtz94    Liveness probe failed: HTTP probe failed with statuscode: 404
44m         Warning   Unhealthy           pod/go-app-deployment-85fbb9ff74-hr2c7    Liveness probe failed: HTTP probe failed with statuscode: 404
44m         Normal    Killing             pod/go-app-deployment-85fbb9ff74-xtz94    Container go-app failed liveness probe, will be restarted
44m         Normal    Killing             pod/go-app-deployment-85fbb9ff74-hr2c7    Container go-app failed liveness probe, will be restarted
43m         Warning   InspectFailed       pod/go-app-deployment-59668f7968-nzg8n    Failed to apply default image tag "ghcr.io/NightFury2283/containers-lab-1/app:new-tag": couldn't par
e image name "ghcr.io/NightFury2283/containers-lab-1/app:new-tag": invalid reference format: repository name (NightFury2283/containers-lab-1/app) must be lowercase
43m         Warning   BackOff             pod/go-app-deployment-85fbb9ff74-dxc7b    Back-off restarting failed container go-app in pod go-app-deployment-85fbb9ff74-dxc7b_default(11987b
0-f1f3-41ba-b8f5-cb74bd113082)
42m         Normal    Pulled              pod/go-lr1-pod                            Container image "ghcr.io/nightfury2283/containers-lab-1/app:sha-c94b5e6" already present on machine
nd can be accessed by the pod
40m         Warning   BackOff             pod/go-lr1-pod                            Back-off restarting failed container go-lr1 in pod go-lr1-pod_default(da8ccc73-e938-4f74-bffc-89c79d
87513)
39m         Normal    Pulled              pod/go-app-deployment-85fbb9ff74-dxc7b    Container image "ghcr.io/nightfury2283/containers-lab-1/app:sha-c94b5e6" already present on machine
nd can be accessed by the pod
39m         Normal    Pulled              pod/go-app-deployment-85fbb9ff74-xtz94    Container image "ghcr.io/nightfury2283/containers-lab-1/app:sha-c94b5e6" already present on machine
nd can be accessed by the pod
39m         Warning   Unhealthy           pod/go-app-deployment-85fbb9ff74-xtz94    Readiness probe failed: HTTP probe failed with statuscode: 404
38m         Normal    Killing             pod/postgres-deployment-5bcd58fdd7-c5xn9  Stopping container postgres
38m         Normal    Killing             pod/nginx-replicaset-9j57c                Stopping container nginx
```

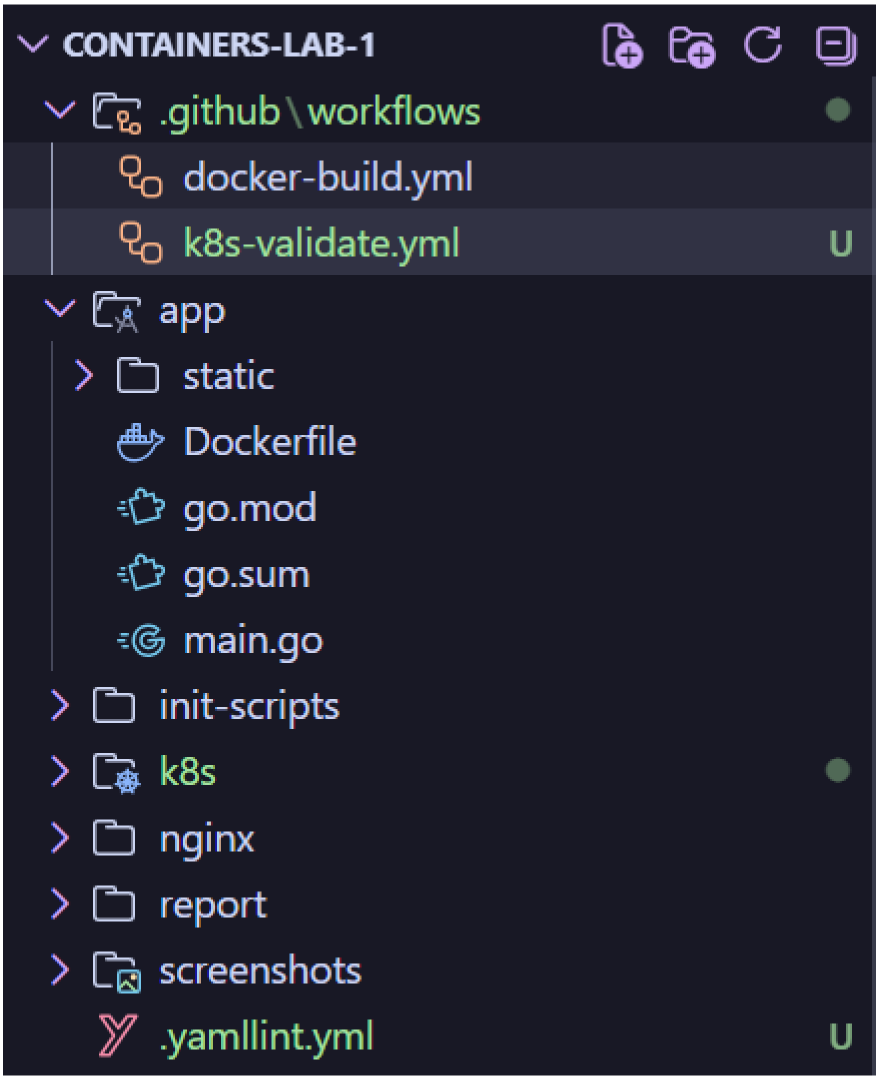
```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs -l app=go-app
2026/04/01 11:08:16 Server starting on port 8080
2026/04/01 11:11:04 Server starting on port 8080

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods
NAME                                  READY   STATUS             RESTARTS        AGE
go-app-deployment-85fbb9ff74-rvghb    0/1     CrashLoopBackOff   12 (3m3s ago)   35m
go-app-deployment-85fbb9ff74-s5mmf    0/1     Running            1 (17s ago)     81s
postgres-deployment-5bcd58fdd7-kx5wv  1/1     Running            0               28m

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs go-app-deployment-85fbb9ff74-rvghb --previous
2026/04/01 11:08:16 Server starting on port 8080

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl port-forward pod/go-app-deployment-85fbb9ff74-rvghb 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
```

```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl run debug --image=nicolaka/netshoot:latest -it --rm -- bash
If you don't see a command prompt, try pressing enter.
debug:~# curl go-app-service:8080/health
curl: (7) Failed to connect to go-app-service port 8080 after 2 ms: Could not connect to server
debug:~# nslookup postgres-service
;; Got recursion not available from 10.96.0.10
Server:         10.96.0.10
Address:        10.96.0.10#53

Name:   postgres-service.default.svc.cluster.local
Address: 10.108.246.120
;; Got recursion not available from 10.96.0.10

debug:~# exit
exit
Session ended, resume using 'kubectl attach debug -c debug -i -t' command when the pod is running
pod "debug" deleted

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$
```

3.3 Мониторинг через Dashboard

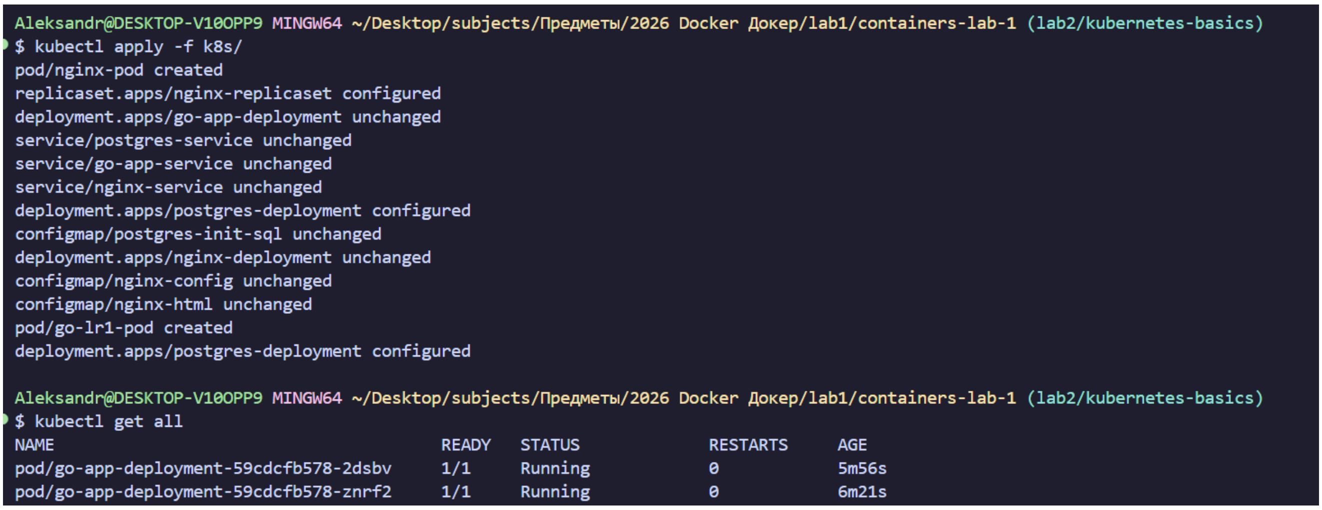


Часть 4. GitHub Actions для проверки манифестов



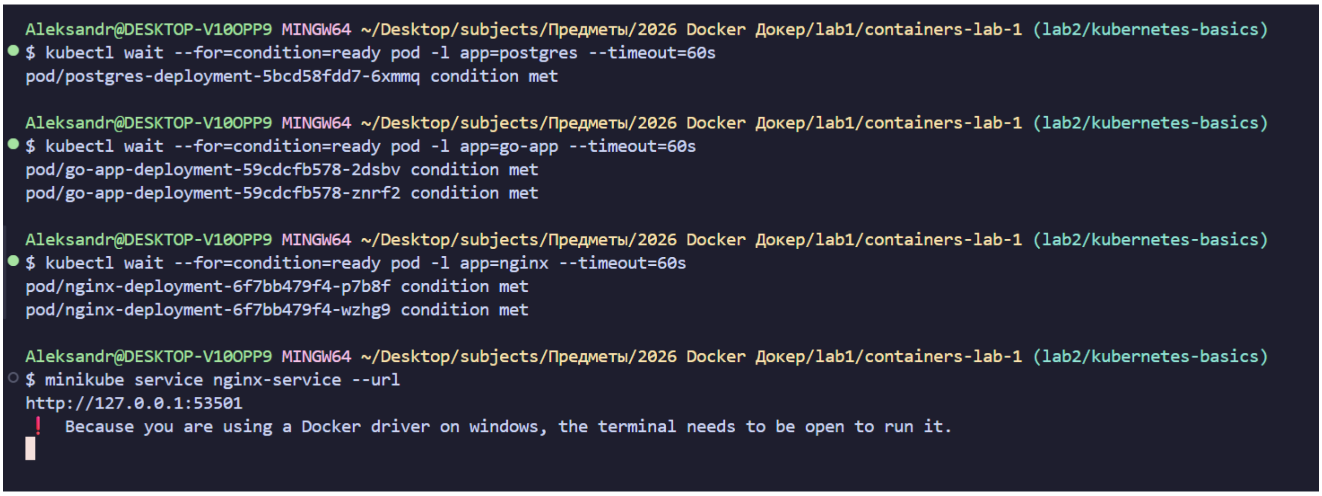
Часть 5. Финальный запуск и проверка

5.1 Главная страница приложения

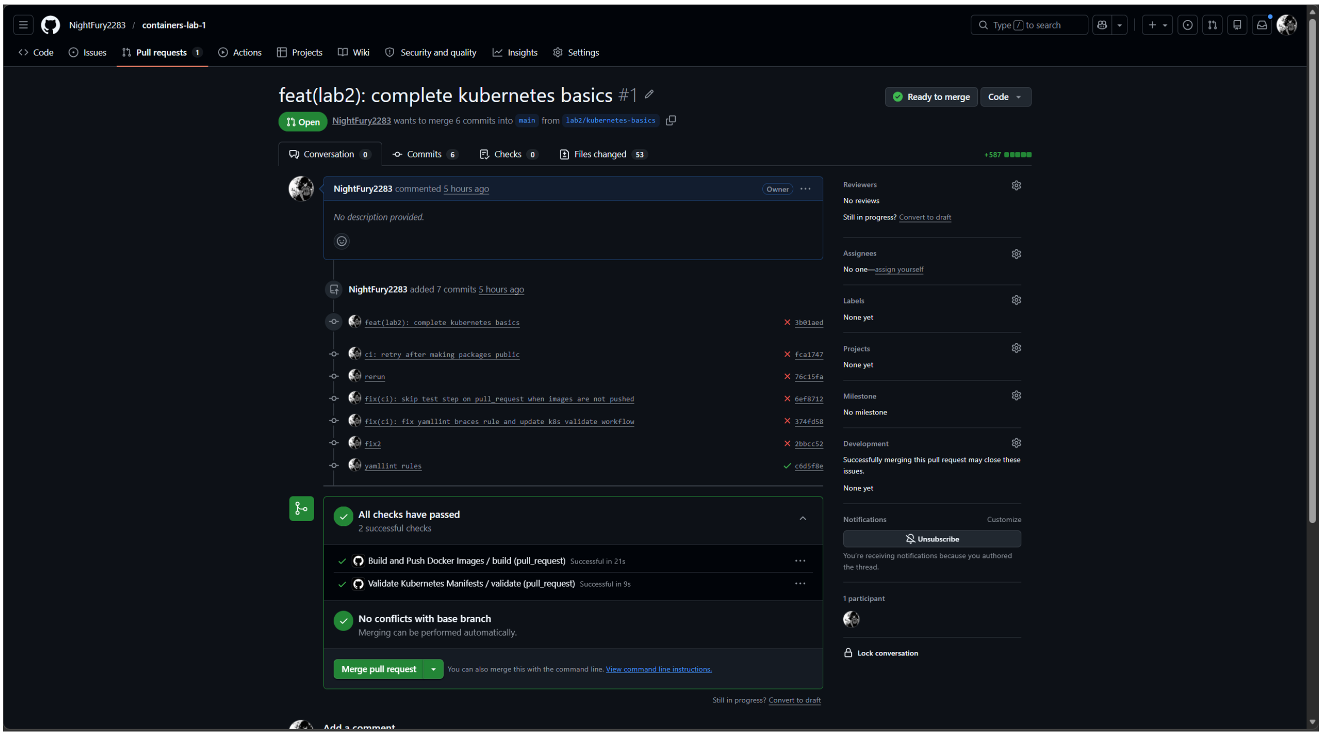


```
Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/
pod/nginx-pod created
replicaset.apps/nginx-replicaset configured
deployment.apps/go-app-deployment unchanged
service/postgres-service unchanged
service/go-app-service unchanged
service/nginx-service unchanged
deployment.apps/postgres-deployment configured
configmap/postgres-init-sql unchanged
deployment.apps/nginx-deployment unchanged
configmap/nginx-config unchanged
configmap/nginx-html unchanged
pod/go-lr1-pod created
deployment.apps/postgres-deployment configured

Aleksandr@DESKTOP-V10OPP9 MINGW64 ~/Desktop/subjects/Предметы/2026 Docker Докер/lab1/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get all
NAME                                  READY   STATUS    RESTARTS   AGE
pod/go-app-deployment-59cdcfb578-2dsbv   1/1     Running   0          5m56s
pod/go-app-deployment-59cdcfb578-znrf2   1/1     Running   0          6m21s
```

5.2 Результат GET /api/users



5.3 GitHub Actions CI (Зеленая галочка)



Часть 6. Ответы на контрольные вопросы

1. **В чем разница между Pod и Deployment?**
   Pod — это минимальная единица развертывания в K8s (один или несколько контейнеров). Deployment — это контроллер более высокого уровня, который управляет ReplicaSet и обеспечивает стратегии обновления (RollingUpdate), отката и автоматического масштабирования.

2. **Для чего нужен Service типа ClusterIP?**
   Service типа ClusterIP предоставляет стабильный внутренний IP-адрес и DNS-имя группе подов внутри кластера. Он позволяет компонентам системы (например, бэкенду обращаться к БД) находить друг друга по имени сервиса, даже если поды пересоздаются и меняют свои внутренние IP.

3. **Как ReplicaSet обеспечивает самовосстановление?**
   ReplicaSet постоянно отслеживает текущее количество работающих подов и сравнивает его с заданным значением `replicas`. Если под падает или удаляется, ReplicaSet мгновенно создает новый на основе шаблона `PodTemplate`, чтобы поддерживать желаемое состояние.

4. **Что произойдет с приложением, если удалить под PostgreSQL?**
   Deployment автоматически перезапустит под PostgreSQL. Однако, так как в данной работе мы использовали `emptyDir` (временное хранилище), все данные в базе данных будут потеряны. Для сохранения данных в реальных проектах используются `PersistentVolumes`.

Часть 7. Выводы

В ходе практической работы был успешно развернут локальный кластер Kubernetes с использованием Minikube. Освоены основные концепции оркестрации: управление жизненным циклом приложений через Deployment, балансировка нагрузки через Service и хранение конфигураций в ConfigMap. Была настроена автоматизированная валидация манифестов в GitHub Actions, что гарантирует корректность YAML-файлов перед деплоем. Самой интересной частью было решение проблем со статус-пробами (Readiness/Liveness probes) и настройка сетевого взаимодействия между Go-приложением и базой данных PostgreSQL.